

STREAMING EVENT ANOMALY DETECTOR

IMPLEMENTATION PLAN

Project Overview

The Streaming Event Anomaly Detector is an AI-powered real-time monitoring system designed to process continuous event streams, detect anomalies using statistical techniques, generate AI-based explanations, classify severity levels, and trigger automated notifications.

The project integrates FastAPI, React, SQLite, Ollama (Llama 3.1), JWT Authentication, and Discord alerting to provide a complete end-to-end anomaly detection platform.

Project Goal

To develop a scalable monitoring platform capable of:

- Generating and processing streaming events
- Detecting anomalies in real time
- Providing AI-generated explanations
- Performing intelligent decision making
- Triggering automated alerts
- Visualizing operational metrics through dashboards

PROJECT PHASES

Phase 1

Project Setup & Environment Configuration

Objectives

- Create backend project structure
- Create frontend project structure
- Configure development environment
- Initialize Git repository
- Setup Docker configuration
- Configure environment variables

Activities

Backend Setup

- FastAPI installation
- Dependency configuration
- API structure creation

Frontend Setup

- React installation
- Vite configuration
- Dashboard initialization

Repository Setup

- Git initialization
- Branch management
- Version control configuration

Deliverables

✓ Backend created

✓ Frontend created

✓ Git repository configured

✓ Environment variables configured

Phase 2

Database Design & Configuration

Objectives

- Design database schema
- Create database tables
- Configure SQLite
- Implement ORM models

Activities

User Table

Stores authentication information.

Event Table

Stores incoming event records.

Anomaly Table

Stores anomaly detection results.

Agent Decision Table

Stores AI recommendations.

Deliverables

- ✓ Database schema finalized
- ✓ Tables created
- ✓ Database connection verified

Phase 3

Authentication & Authorization

Objectives

- Implement JWT Authentication
- Create login APIs
- Create registration APIs
- Implement role management

Activities

User Registration

Create user accounts.

User Login

Generate JWT tokens.

Protected Routes

Restrict unauthorized access.

Deliverables

- ✓ Login implemented
- ✓ Registration implemented
- ✓ JWT security enabled
- ✓ Protected endpoints operational

Phase 4

Event Generation Module

Objectives

- Simulate real-time event streams
- Generate test events continuously
- Produce normal and abnormal events

Activities

Event Producer

Creates streaming event records.

Event Generator Logic

Generates randomized event values.

Event Scheduling

Creates continuous event flow.

Deliverables

- ✓ Event generation operational
- ✓ Continuous stream simulation completed

Phase 5

Event Queue & Processing Engine

Objectives

- Implement event queue
- Process events continuously
- Ensure event delivery

Activities

Queue Management

Store events temporarily.

Event Consumer

Read events from queue.

Processing Pipeline

Forward events to anomaly detector.

Deliverables

- ✓ Queue implementation completed
- ✓ Consumer operational
- ✓ Event processing functional

Phase 6

Anomaly Detection Engine

Objectives

- Detect abnormal events
- Calculate statistical deviations
- Generate anomaly records

Activities

Z-Score Calculation

Compute deviation from normal behavior.

Threshold Evaluation

Compare values against threshold.

Anomaly Classification

Identify suspicious events.

Deliverables

- ✓ Detection engine completed
- ✓ Threshold validation operational
- ✓ Anomaly classification working

Phase 7

AI Explanation Engine

Objectives

- Integrate Ollama
- Use Llama 3.1
- Generate anomaly explanations

Activities

AI Integration

Connect Ollama service.

Prompt Engineering

Generate contextual explanations.

Explanation Storage

Save AI responses.

Deliverables

- ✓ Ollama integrated
- ✓ AI explanations generated
- ✓ Results persisted

Phase 8

AI Agent Decision System

Objectives

- Perform intelligent analysis
- Determine severity
- Recommend actions

Activities

Severity Assessment

Classify:

- LOW
- MEDIUM
- HIGH
- CRITICAL

Recommendation Generation

Suggest corrective actions.

Alert Decision Logic

Determine if alerts are necessary.

Deliverables

- ✓ AI Agent operational
- ✓ Severity classification implemented
- ✓ Recommendations generated

Phase 9

Alerting & Notification System

Objectives

- Deliver real-time alerts
- Notify stakeholders

Activities

Discord Integration

Configure webhooks.

Alert Formatting

Generate readable messages.

Critical Event Handling

Send immediate notifications.

Deliverables

- ✓ Discord integration completed
- ✓ Alert delivery verified
- ✓ Critical notifications operational

Phase 10

Dashboard Development

Objectives

- Build monitoring dashboard
- Visualize event metrics
- Display anomalies

Activities

KPI Cards

- Total Events
- Total Anomalies
- Active Alerts
- Processing Rate

Charts

- Event Timeline
- Severity Distribution
- Alert Trends

Tables

- Event Logs
- Anomaly Records

Deliverables

- ✓ Dashboard completed
- ✓ Charts functional
- ✓ Real-time monitoring available

Phase 11

Testing & Quality Assurance

Objectives

- Validate all modules
- Identify defects
- Improve reliability

Activities

Functional Testing

Verify system functionality.

Security Testing

Validate authentication.

Performance Testing

Stress-test event processing.

Integration Testing

Validate module interactions.

Deliverables

- ✓ Test cases executed

✓ Security validated

✓ Performance verified

✓ Defects resolved

Phase 12

Dockerization & Deployment

Objectives

- Containerize application
- Prepare production deployment

Activities

Docker Setup

Create Docker configuration.

Container Testing

Validate deployment.

Environment Management

Configure runtime variables.

Deliverables

✓ Docker image created

✓ Deployment tested

✓ Runtime environment validated

Phase 13

Documentation & Final Submission

Objectives

- Prepare technical documentation
- Prepare project report
- Prepare presentation materials

Activities

Documentation

- PRD
- TRD
- App Flow
- Backend Schema
- Test Cases

Presentation

- PPT Preparation
- Demo Script
- Architecture Diagram

Final Review

- Code Review
- Documentation Review
- Submission Validation

Deliverables

- ✓ Documentation completed
- ✓ PPT completed
- ✓ Demo prepared
- ✓ Project submission ready

Project Timeline

Week	Activities
-------------	-------------------

Week 1	Setup & Environment Configuration
--------	-----------------------------------

Week 2	Database Design
--------	-----------------

Week 3	Authentication
--------	----------------

Week 4	Event Generator & Queue
--------	-------------------------

Week 5	Anomaly Detection Engine
--------	--------------------------

Week 6	AI Explanation Engine
--------	-----------------------

Week 7	AI Agent
--------	----------

Week 8	Alert System
--------	--------------

Week 9	Dashboard Development
--------	-----------------------

Week 10	Testing
---------	---------

Week 11	Docker Deployment
---------	-------------------

Week 12	Documentation & Submission
---------	----------------------------

Success Criteria

✓ Events generated successfully

✓ Events processed continuously

✓ Anomalies detected accurately

✓ AI explanations generated

✓ Severity classification working

✓ Discord alerts delivered

✓ Dashboard monitoring functional

✓ Authentication secured

✓ Database persistence verified

✓ Docker deployment completed

Conclusion

The implementation plan provides a structured roadmap for developing the Streaming Event Anomaly Detector. The phased approach ensures systematic development, testing, deployment, and documentation of all project modules while maintaining scalability, security, and reliability throughout the project lifecycle.